

Pornography Consumption, Trait Self-Control, and Daily Productivity Among Young Adults: A Cross-Sectional Mediation and Developmental Analysis

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Abstract

This study investigated the relationships between pornography consumption, self-control, and daily productivity among young adults, with specific focus on the mediating role of self-control, the developmental impact of age of first exposure, and gender differences. A cross-sectional survey design was used, with data collected from 250 young adults aged 18 and above through an anonymous online questionnaire. Pornography consumption frequency, self-control, productivity, age of first exposure, and demographic characteristics were measured. Data were analysed using Spearman's rank-order correlation, bootstrapped mediation analysis (5,000 resamples), one-way ANOVA with Tukey HSD post-hoc tests, Mann-Whitney U tests, and Welch's t-tests at $\alpha = 0.05$. Results revealed a statistically significant, weak negative correlation between pornography consumption and productivity ($r_s = -0.157$, $p = 0.013$). Self-control fully mediated this relationship, with a significant indirect effect ($\beta = -0.188$, 95% CI [-0.287, -0.098]) and a non-significant direct effect ($p = 0.681$) when the mediator was included. Age of first exposure was significantly associated with self-control ($F = 4.12$, $p = 0.003$) and productivity ($F = 3.23$, $p = 0.013$), with earlier exposure predicting lower outcomes; however, it did not significantly predict current pornography consumption ($F = 1.52$, $p = 0.198$). Notably, the majority of first exposures were not actively sought, with accidental online discovery being the most common pathway (27.2%). Males reported significantly higher pornography consumption than females ($p < 0.001$), but no significant gender differences were found in self-control or productivity. These findings suggest that pornography consumption affects productivity indirectly through its erosion of self-control, supporting ego depletion theory. The results have implications for interventions targeting self-regulation, particularly during early adolescence, and highlight the need for longitudinal research to establish causality.

Keywords: pornography consumption, self-control, productivity, ego depletion, young adults, mediation analysis

INTRODUCTION

In past generations, access to sexually explicit material often required individuals to actively seek it through relatively limited and less immediate channels. The availability of reliable, high-speed internet and smartphones has transformed how people access information and digital media, bringing significant advantages in communication, education, and everyday life. At the same time, these technologies have also made sexually explicit material more readily accessible, allowing pornography to be encountered and viewed with considerably less effort than in the past. The data from the present study indicate that the majority of respondents first exposure to pornography was not intentional, occurring instead through accidental online discovery, traditional media, or exposure by others. This suggests that the digital and media environment itself through sexualised advertising, unregulated platforms and even algorithm driven content, plays a huge role in initiating exposure.

Daily productivity represents another potentially relevant outcome. For young adults, the ability to regulate attention and behavior is important for maintaining work, study, and other goal-directed activities. It is therefore plausible that frequent engagement with highly reinforcing online content could be associated with differences in perceived productivity.

Two findings about pornography consumption are well established: problematic use is characterised by impaired control (Büschel et al., 2022), and it is associated with cognitive deficits including impaired inhibitory control and poorer working memory (Castro-Calvo et al., 2021; Müller et al., 2025). What remains uncertain is the consequence for everyday functioning. Tokan et al. (2025) found that students addicted to pornography had productivity scores 1.30 points lower than non-addicted students, and Tewogbola et al. (2025) reported a 2.49-point deficit on an academic success inventory. These studies establish that an association exists but do not explain why. Two mechanisms have been proposed; time displacement and cognitive interference, yet neither has been formally tested as a mediator.

The timing of first exposure may provide an additional perspective on these relationships. Exposure to pornography can occur at different stages of development, and individuals who first encounter sexually explicit material during adolescence may differ from those whose first intentional exposure occurs during adulthood. Research involving younger populations has examined associations between online pornography use and psychological outcomes, although the evidence remains heterogeneous and does not by itself establish that the timing of exposure causes later differences in psychological or behavioral functioning.

Gender may also be relevant when examining pornography consumption and related behavioral characteristics. Previous research has frequently examined gender differences in pornography use, but differences in consumption frequency do not necessarily imply corresponding differences in self-control or productivity. Moreover, evidence concerning problematic use and its psychological correlates can vary according to how pornography-related problems are defined and measured.

Despite the growing literature on pornography use, self-regulation, and related psychological outcomes, an important empirical question remains: to what extent is the association between pornography consumption and everyday productivity statistically related to individual differences in self-control? In addition, less attention has been given to how self-control and perceived productivity vary according to age at first intentional exposure, particularly when exposure is examined across several developmental age groups. Examining gender differences in consumption, self-control, and productivity may provide further context for interpreting these relationships.

The present study addresses these gaps through a cross-sectional survey of 250 adults aged 18 and above, recruited from multiple online platforms and representing a culturally diverse population. Participants completed measures of pornography consumption frequency, self-

control (five-item composite), self-rated productivity, stress, age of first exposure, and demographics. Four analytical approaches were used: Spearman's rank-order correlation to test the pornography–productivity relationship; bootstrapped mediation analysis (5,000 resamples) to test self-control as a mediator; one-way ANOVA with Tukey HSD post-hoc tests to examine age-of-exposure effects; and Mann-Whitney U and Welch's t-tests to compare genders. By testing the mechanism, examining developmental timing, and comparing outcomes across genders, the study moves beyond documenting an association to explaining it.

Aim and Objectives

The aim of this study is to investigate the relationships between pornography consumption, self-control, and daily productivity among young adults. Specifically, the study examines the mediating role of self-control, the developmental impact of age of first exposure, and the gender differences that shape these variables. Four objectives guide the investigation:

1. To determine whether pornography consumption is significantly correlated with productivity among young adults.
2. To test whether self-control mediates the relationship between pornography consumption and productivity.
3. To examine whether age of first exposure is associated with current pornography consumption, self-control, and productivity among young adults.
4. To compare pornography consumption, self-control, and productivity between male and female young adults.

Hypotheses

The following hypotheses were tested:

H₁: There is a significant relationship between pornography consumption and productivity among young adults.

H₂: Self-control significantly mediates the relationship between pornography consumption and productivity.

H₃: Age of first exposure is significantly associated with current pornography consumption, self-control, or productivity.

H₄: There are significant gender differences in pornography consumption, self-control, or productivity.

METHODS

Research Design

This study employed a cross-sectional survey design. A quantitative approach was adopted, using self-report measures to collect numerical data suitable for statistical analysis.

Participants

The target population was young adults aged 18 and above. A total of 250 valid responses were collected. Participants were recruited using convenience and snowball sampling via multiple online platforms, including SurveyCircle, SurveySwap, LinkedIn, and personal networks. The survey was restricted to adults aged 18 and above. No personally identifiable information was collected. Data collection took place between 25th June 2026 and 8th September 2026.

The demographic characteristics of the sample are presented in Table 1.

Table 1: Demographic Characteristics of Participants (N = 250)

Variable	Category	n	%
Age	18–20	37	14.8
	21–24	99	39.6
	25–29	63	25.2
	30–34	22	8.8
	35+	29	11.6
Gender	Male	113	45.2
	Female	136	54.4
	Prefer not to say	1	0.4
Relationship Status	Single (not dating)	129	51.6
	Casually dating	28	11.2
	In a committed relationship	64	25.6
	Married	20	8.0
	Prefer not to say	9	3.6
Employment Status	Student	147	62.6
	Employed	66	28.1
	Self-employed	13	5.5
	Unemployed	9	3.8
Region	Europe	87	34.8
	Asia	59	23.6
	Other / Not Specified	47	18.8
	Africa	21	8.4

	North America	19	7.6
	South America	10	4.0
	Oceania	7	2.8

Measures

Pornography Consumption.

Frequency of intentional viewing over the past 30 days was measured on an ordinal scale ranging from 0 days to 22–30 days. Responses were coded ordinally from 0 to 5.

Self-Control.

Measured using five Likert-scale items adapted from the Brief Self-Control Scale (1 = Strongly Disagree to 5 = Strongly Agree). Three negatively worded items were reverse-coded so that higher scores reflected higher self-control. The composite score was calculated as the mean of the five items. Internal consistency was acceptable (Cronbach's $\alpha = 0.743$).

Productivity.

Self-rated productivity over the past seven days, measured on a single item scale from 1 (Extremely unproductive) to 10 (Extremely productive).

Age of First Exposure.

Respondents reported the age at which they first intentionally viewed pornography. Responses were coded ordinally from 1 (under 10 years) to 5 (22+ years).

Demographics.

Age, gender, relationship status, education, employment status, and nationality were collected for descriptive purposes. Only gender was used as a grouping variable in hypothesis testing.

Procedure

The survey was administered via Google Forms. The questionnaire began with a consent statement informing participants that the survey was anonymous and intended for adults aged 18 and above. The instrument consisted of five sections: productivity (5 items), self-control (5 items), pornography viewing habits (9 items), demographics (6 items), and an optional email field for follow-up. No personally identifiable information was collected.

Data Analysis

Data were analysed using Python with the pandas, numpy, scipy, statsmodels, and pingouin libraries. A significance level of $\alpha = 0.05$ was adopted throughout. Descriptive statistics (means, standard deviations, frequencies) were calculated to summarise the sample. Spearman's rank-order correlation was used to test the relationship between pornography consumption and productivity. A bootstrapped mediation analysis (5,000 resamples) tested whether self-control mediated this relationship. One-way ANOVA with Tukey HSD post-hoc tests examined

differences across age-of-exposure groups. Mann-Whitney U tests and Welch's t-tests compared males and females on all key variables.

RESULTS

Descriptive Statistics

The final sample consisted of 250 young adults. Descriptive statistics for key variables are presented in Table 2.

Table 2: Descriptive statistics for key variables

Variable	M	SD	1	2	3
1. Pornography Frequency	1.38	1.48	—		
2. Self-Control	3.09	0.76	-0.260*	—	
3. Productivity	6.11	1.92	-0.157*	0.425*	—

Note: All correlations are Spearman's rho. N = 250 (self-control N = 235 due to missing items).

**p < 0.05*

Of the 250 respondents, 86 (34.4%) reported no intentional pornography viewing in the past 30 days, while 164 (65.6%) reported some level of consumption. Regarding age of first exposure, 174 respondents (74.0% of the 235 with valid data) were first exposed before age 18. Accidental online discovery was the most common pathway to first exposure (n = 68, 27.2%), followed by intentional search (n = 52, 20.8%), and introduction by a friend or peer (n = 50, 20.0%).

Objective 1: Correlation Between Pornography Consumption and Productivity

A Spearman's rank-order correlation revealed a statistically significant, weak negative correlation between pornography consumption frequency and productivity, $r_s = -0.157$, $p = 0.013$. Higher pornography consumption was associated with slightly lower productivity.

The null hypothesis stated that there is no relationship between pornography consumption and productivity. Because the p-value was below the 0.05 threshold, the null hypothesis is rejected. The alternative hypothesis is supported: a statistically significant, negative relationship exists between the two variables.

Objective 2: Mediation Analysis

A bootstrapped mediation analysis (5,000 resamples) tested whether self-control mediates the relationship between pornography consumption and productivity. Results are presented in Table 3.

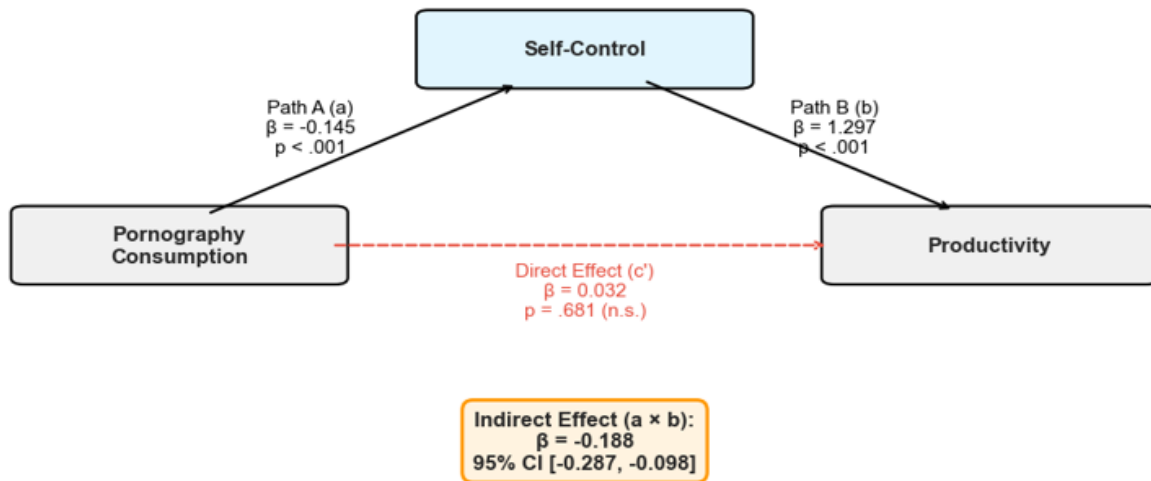
Table 3: Mediation Analysis Results

Effect	β	SE	p-value	95% CI
Path A (Porn $\rightarrow$ Self-Control)	-0.145	0.04	< 0.001	—
Path B (Self-Control $\rightarrow$ Productivity)	1.297	0.18	< 0.001	—
Direct Effect (c')	0.032	0.08	0.681	—
Indirect Effect (a $\times$ b)	-0.188	—	—	[-0.287, -0.098]

Note: The indirect effect was tested using bootstrapped 95% confidence intervals (5,000 resamples). Because the sampling distribution of the product of coefficients is not normally distributed, a traditional p-value is not appropriate for the indirect effect.

Path A was significant: higher pornography consumption predicted lower self-control. Path B was also significant: higher self-control predicted higher productivity. The indirect effect was significant (95% CI did not include zero), while the direct effect became non-significant when self-control was included. This pattern indicates full mediation: pornography consumption affects productivity through its impact on self-control.

The null hypothesis stated that self-control does not mediate the relationship between pornography consumption and productivity. Because the indirect effect was statistically significant, the null hypothesis is rejected.

**Fig 1: Mediation Analysis.**

Objective 3: Age of First Exposure

A one-way ANOVA examined whether age of first exposure was associated with current self-control and productivity. Results are presented in Table 3.

Table 4: One-Way ANOVA Results by Age of First Exposure

Outcome	df (between, within)	F	p
Self-Control	(4, 218)	4.12	0.003
Productivity	(4, 218)	3.23	0.013
Pornography Consumption	(4, 214)	1.52	0.198

The ANOVA revealed statistically significant differences across exposure groups for both self-control ($F = 4.12$, $p = 0.003$) and productivity ($F = 3.23$, $p = 0.013$). For pornography consumption, the group means displayed a consistent downward trend (under-10 group $M = 1.79$; 22+ group $M = 1.00$), but the result did not reach statistical significance ($F = 1.52$, $p = 0.198$).

Tukey HSD post-hoc tests identified specific group differences for self-control. Results are presented in Table 5.

Table 5: Tukey HSD post-hoc tests for self-control.

Outcome	Group 1	Group 2	Mean Difference	Adjusted p
Self-Control	11–13 years	18–21 years	0.53	0.003
Self-Control	14–17 years	18–21 years	0.48	0.014

The null hypothesis stated that age of first exposure is not associated with current pornography consumption, self-control, or productivity. Because the ANOVA revealed significant differences for self-control and productivity, therefore, the null hypothesis is rejected for self-control and productivity but not for current pornography consumption.

Objective 4: Gender Differences

Independent samples tests compared males and females on all three key variables. Given the ordinal nature of the pornography frequency variable and the unequal variances observed, a Mann-Whitney U test was used for consumption frequency, while Welch's t-tests were applied to self-control and productivity. Results are presented in Table 6.

Table 6: Gender Differences in Key Variables

Variable	Female M (SD)	Male M (SD)	Test	p-value	Effect Size
Pornography Frequency	0.85 (1.00)	2.04 (1.70)	Mann-Whitney U	< 0.001	0.41
Self-Control	3.12 (0.75)	3.05 (0.77)	Welch's t	0.485	0.09
Productivity	5.94 (1.91)	6.28 (1.93)	Welch's t	0.162	-0.18

Note: Female n = 136; Male n = 113. Self-control N was 127 females and 107 males due to missing items. One respondent (Prefer not to say) was excluded.

Males reported significantly higher pornography consumption than females ($p < 0.001$, effect size = 0.41). However, no significant gender differences were found in self-control ($p = 0.485$) or productivity ($p = 0.162$).

The null hypothesis stated that there are no significant differences between males and females in pornography consumption, self-control, or productivity. The null hypothesis is rejected for pornography consumption but not rejected for self-control or productivity.

DISCUSSION

Self-control fully mediates the relationship between pornography consumption and productivity. Pornography consumption does not directly lower productivity; instead, it reduces self-control, which in turn lowers productivity. This finding supports ego depletion theory (Baumeister & Tierney, 2011) and is consistent with Büsche et al. (2022), who identified self-regulation failure as a core feature of problematic pornography use, and with Schiebener et al. (2015), who found that individuals with cybersex addiction showed reduced executive control in multitasking situations.

The correlation between pornography consumption and productivity firmly establish that pornography consumption is negatively associated daily productivity ($r_s = -0.157$, $p = 0.013$). While the small correlation magnitude suggests that pornography is not the primary driver of productivity differences. This aligns with Tokan et al. (2025) and Tewogbola et al. (2025).

The finding that age of first exposure was significantly associated with both current self-control ($F = 4.12$, $p = 0.003$) and productivity ($F = 3.23$, $p = 0.013$) suggests that the timing of first exposure may have a lasting developmental impact. Individuals first exposed during early adolescence (11–17 years) reported significantly lower self-control in adulthood than those first exposed at 18–21 years. This pattern aligns with the view that adolescence is a critical period for the development of executive functions such as impulse control and decision-making (Ugese et al., 2024). Exposure during this window may disrupt the formation of self-regulatory habits, with

consequences that persist into adulthood. However, the association between age of first exposure and current pornography consumption was not statistically significant ($F = 1.52$, $p = 0.198$). This indicates that age of first exposure does not inevitably result in higher consumption frequency in adulthood.

The finding that males reported significantly higher pornography consumption than females ($M = 2.04$ vs. 0.85 , $p < 0.001$) is consistent with the broader literature (Tokan et al., 2025; Ugees et al., 2024). However, the absence of significant gender differences in either self-control ($p = 0.485$) or productivity ($p = 0.162$) is more notable. It suggests that despite viewing pornography at more than double the rate of females, males do not suffer lower self-control or reduced productivity compared to females. This suggests that higher male consumption rates do not create a gender disparity in executive function or daily performance, as both sexes exhibit comparable self-regulatory and productivity outcomes.

CONCLUSION

This study demonstrates that pornography consumption affects productivity indirectly, through its impact on self-control. Age of first exposure emerged as a significant developmental factor, with early exposure predicting lower self-control and productivity in adulthood, and with most first exposures occurring incidentally rather than through intentional search. However, the association between age of first exposure and current pornography consumption was not statistically significant indicating that age of first exposure does not inevitably result in higher consumption frequency in adulthood.

Gender differences were observed in consumption but not in outcomes as males reported significantly higher pornography use than females, yet no significant differences were found in self-control or productivity. This suggests that the mechanism linking pornography to productivity operates similarly across genders and is not male specific. While the relationship between pornography consumption and productivity is statistically significant, its weak magnitude indicates that pornography is one of many factors influencing productivity. Future research should employ longitudinal designs to establish causality and examine these relationships in more diverse cultural contexts.

REFERENCES

Baumeister, R. F., & Tierney, J. (2011). *Willpower: Rediscovering the greatest human strength*. Penguin Press.

Büsche, K., Stark, R., Brand, M., & Antons, S. (2022). Self-regulatory processes in problematic pornography use. *Current Addiction Reports*, 9, 344–352.

Castro-Calvo, J., Cervigón-Carrasco, V., Ballester-Arnal, R., & Giménez-García, C. (2021). Cognitive processes related to problematic pornography use (PPU): A systematic review of experimental studies. *Addictive Behaviors Reports*, 13, 100345.

Harun, H., et al. (2025). A comprehensive review of pornography in the digital age: Psychological and technological dimensions. *Journal of Behavioral Addictions*.

López-Pinar, C., et al. (2025). Psychotherapy for problematic pornography use: A systematic review and meta-analysis. *Journal of Behavioral Addictions*.

Mestre-Bach, G., & Potenza, M. N. (2025). Adolescent pornography use: A systematic review of longitudinal studies. *Journal of the Korean Academy of Child and Adolescent Psychiatry*.

Müller, S. M., et al. (2025). Self-control abilities in specific types of problematic usage of the internet: Findings from clinically validated samples with neurocognitive tasks. *American Journal of Psychiatry*, 182(7), 660–670.

National Institute on Drug Abuse. (2026). *Adolescent Brain Cognitive Development (ABCD) Study: Psychosocial correlates of early pornography exposure*. Research Square.

Schiebener, J., Laier, C., & Brand, M. (2015). Getting stuck with pornography? Overuse or neglect of cybersex cues in a multitasking situation is related to symptoms of cybersex addiction. *Journal of Behavioral Addictions*, 4(1), 14–21.

Tewogbola, P., Bodunde, V., & Odaudu, D. (2025). Pornography, problematic use and academic outcomes in college students: A cross-sectional analysis. *Journal of Graduate Education Research*, 6(1), 23–35.

Tokan, F. A., Kurniawan, L. S., Ardani, I. G. A. I., Ariani, N. K. P., & Marvella. (2025). Pornography use and its impact on professional performance of Faculty of Medicine Udayana University students. *Jurnal Psikiatri Surabaya*, 14(1), 72–80.

Ugese, I. J., Chinonso, U., Gandi, J. C., et al. (2024). Pornography addiction in the emerging adults: The role of social isolation, self-control and stress coping. *British Journal of Psychology Research*, 12(1), 26–36.